Expense Tracker Testing & Evaluation Guide

A comprehensive guide to verifying the Full-Stack Application

1. Environment Setup & Execution

Starting the Database

Ensure you have MongoDB Community Server installed and running locally on port 27017.

Starting the Backend API

1. Open terminal, navigate to the "backend" directory.
2. Run `npm run dev`. Wait for the console to output "MongoDB Connected: localhost".

Starting the Frontend UI

1. Open a new terminal, navigate to the "frontend" directory.
2. Run `npm run dev`. Open your browser to http://localhost:5173.

2. Frontend UI Testing (http://localhost:5173)

A. Authentication Flow

- Navigate to the app. You will be redirected to the Login page.
- Click "Sign up" at the bottom to register a new account. Submit the form.
- Verify that upon successful registration, you are redirected to the Dashboard.

B. Dashboard & Analytics

- On the Dashboard, observe the "Total Spending", "Active Budgets", and "Alerts".
- Initially, the charts will be empty. Once you add expenses, a Pie Chart and Line Chart will render dynamically.

C. Expense Management

- Navigate to "Expenses" via the sidebar.
- Click "Add Expense", fill out the amount, category, and date, then submit.
- Verify the new expense appears in the table. Click the Trash icon to test deletion.

D. Budget Tracking

- Navigate to "Budgets" via the sidebar.
- Click "Set Budget", specify an amount and category (e.g., $500 for Food).
- Navigate back to the Dashboard. Verify that the "Budget Status" progress bar appears and accurately reflects your spending ratio.

E. Exporting Reports

- Navigate to "Reports" via the sidebar.
- Click "Export as CSV" or "Export as PDF". Verify that a file downloads successfully with your data.

3. Backend API Testing via Swagger (http://localhost:3000/api-docs)

The backend includes an interactive Swagger OpenAPI interface for testing raw endpoints.

A. Getting a JWT Token

- Expand the POST /auth/register or /auth/login endpoint. Click "Try it out".
- Input valid JSON credentials and Execute. Copy the "token" string from the response body.

- Scroll to the top of the page, click the green "Authorize" button, paste your token, and click Authorize.

B. Testing Endpoints

- Expand GET /expenses. Click "Try it out" and "Execute" to fetch all expenses.
- Expand POST /expenses. Input JSON (amount, category) to create an expense.
- Test the GET /analytics/categories and GET /analytics/trends endpoints to verify the MongoDB aggregation pipelines return formatted chart data.

4. Background Jobs & Advanced Features

The backend utilizes `node-cron` to schedule automated tasks in `backend/src/services/cronService.js`.

- The server runs daily checks to evaluate if your actual spending has exceeded 80% of your set budgets.
- If a limit is nearing, the system triggers an email alert using `nodemailer`. (Currently configured for Ethereal dummy SMTP in `.env`).